Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/21/2026
PI Name:	Vanapalli, Siva		
Project Title:	Impact of the Gut Microbiome on the Integrative Physiology of Genetically Diverse Invertebrates		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Lubbock	**State:**	TX
Zip Code:	79409-0001	**Congressional District:**	19
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	03/01/2022	**End Date:**	02/28/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/2027 per F. Hernandez/ARC (Ed., 3/10/26) NOTE: End date changed to 02/28/2026 per B. Stallsmith/ARC (Ed., 5/20/25) NOTE: End date changed to 02/29/2025 per F. Hernandez/ARC (Ed., 4/25/22)		
Key Personnel Changes/Previous PI:	Per the PI: No change from the previous progress report, except that we experienced the tragic loss of co-PI Nathaniel Szewczyk from Ohio University (Ed., 1/21/26).		
COI Name (Institution):	Driscoll, Monica Ph.D. (Rutgers University, New Brunswick) Samuel, Buck S. Ph.D. (Baylor College of Medicine, Inc)		
Grant/Contract No.:	80NSSC22K0250		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The gut microbiome is a complex community of microbes that exert considerable influence over host physiology, development, metabolism, and predisposition to the disease on Earth. Recent spaceflight investigations in both humans and rodents show that the microbiome community structure is altered under microgravity, posing risk factors for crew health. Thus, understanding the impact of the gut microbiome on long-term physiological changes will be crucial for long-duration space missions to Mars and beyond. The current project is centered around addressing two major knowledge gaps in space biology research. First, fundamental understanding is lacking as to how differences in microbiomes contribute to changes in gut colonization, organ-level physiology, and whole-organism function under microgravity. Second, even though it is recognized on Earth that individual genetic variation can have a large impact when organisms within a species are exposed to new environmental conditions, very little knowledge exists on how genetic diversity within individual species impacts the integrative physiology of organisms when exposed to microgravity since most flight studies to date have focused on genetically homogenous rodent models or cell cultures. To address these knowledge gaps, we plan to use C. elegans, which is an established and low-cost invertebrate model for space biology, microbiome studies, and genetic diversity research. We plan to use recently established gut microbial communities to investigate the impact of the gut microbiome on host physiology. In a parallel advance, using genetically diverse wild isolates of C. elegans, we will study host-microbome interactions in spaceflight. Our project plan involves testing the following hypotheses in International Space Station (ISS) flight studies with ground study components: (i) Microbial membership of C. elegans gut influences host transcriptional response, tissue-level physiology, and whole-organism function, (ii) Genetic diversity of host influences gut membership of individual microbes, tissue-level physiology, and whole-organism function, and (iii) Insulin signaling pathway play a central role in driving microbiome-induced host response in spaceflight. The proposed studies are aligned with the stated strategic goals of NASA Space Biology, which defines over-arching guiding questions focused on integrated biological approaches to understand physiological and molecular mechanisms in living systems that respond to space exploration environments. Pre-biotic and probiotic therapies could be potentially realized from our investigations to improve crew health, along with the dissemination of new flight-tested protocols and molecular characterization tools for the spaceflight community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research in this project will quantitate changes in the microbiome as well as in the host in response to space-flight thereby providing detailed understanding of the dynamics of host-microbiome interactions. Additionally, new microfluidic technologies will be developed that facilitate gut-microbiome studies. This new knowledge and capabilities provided by our project can provide significant Earth benefit by contributing to the development of “living medicines” (e.g. probiotics) for treating a variety of human health disorders associated with metabolism, gut, muscle and the nervous system.
Task Progress:	The Driscoll Lab at Rutgers University received frozen samples in polyethylene (PE) bags from the experiment verification testing (EVT) conducted at the Samuel Lab at Baylor College of Medicine. We examined gut morphology, integrity and microbiome occupancy using the ERM-2:::GFP green reporter that visualizes the intestine. These examinations have confirmed that tissue integrity, and gut occupancy (red microbiome) is good for bag culture over the period anticipated to represent flight duration. • In these studies, we have also defined a number of measures we intend to check for post-flight comparison, and we have established protocols for image stitching that enable a whole animal view of the intestine. • We completed the analysis of the touch neuron morphology in Pmec-4::mCherry strains (green microbiome in this case). We defined a list of parameters that can be measured for post-flight comparison, and we find that these features appear as expected for ground controls. • We initiated a detailed quantitation of stress reporter DAF-16:GFP and confirmed that we can image nuclear translocation under stress. • We contributed a book chapter detailing the protocols for spaceflight gut microbiome studies with C. elegans. Blackburn, D. R., Rahman, B., Saroyia, A. P., Parish, A. J., Driscoll, M., Szewczyk, N. J., Vanapalli, S. A., Samuel, B. S. Defining Microbiome Impact on Host Physiology During Spaceflight Using Caenorhabditis elegans. In: Biomedicine and Biotechnology in Space, ed. M. Bizzarri. Springer Nature, 2025. Accepted. [Ed. Note: See Bibliography for complete citation when published.] Launch preparation milestones were completed to originally support the SpaceX CRS-32 (SpX-32) mission, scheduled for April 2025, but the payload was removed from the vehicle due to International Space Station (ISS) constraints. The "C. elegans Biological Investigation on Microbiome Effect in Space" (CBIOMES) project was then manifested for the SpaceX CRS-34 (SpX-34) mission, scheduled for April 2026.
Bibliography Type:	Description: (Last Updated: 10/20/2025)